

Name: _____

Class: _____

Mr. Rawson • Computer Science



Year 10 2026

Pixel Art and Colour Depth

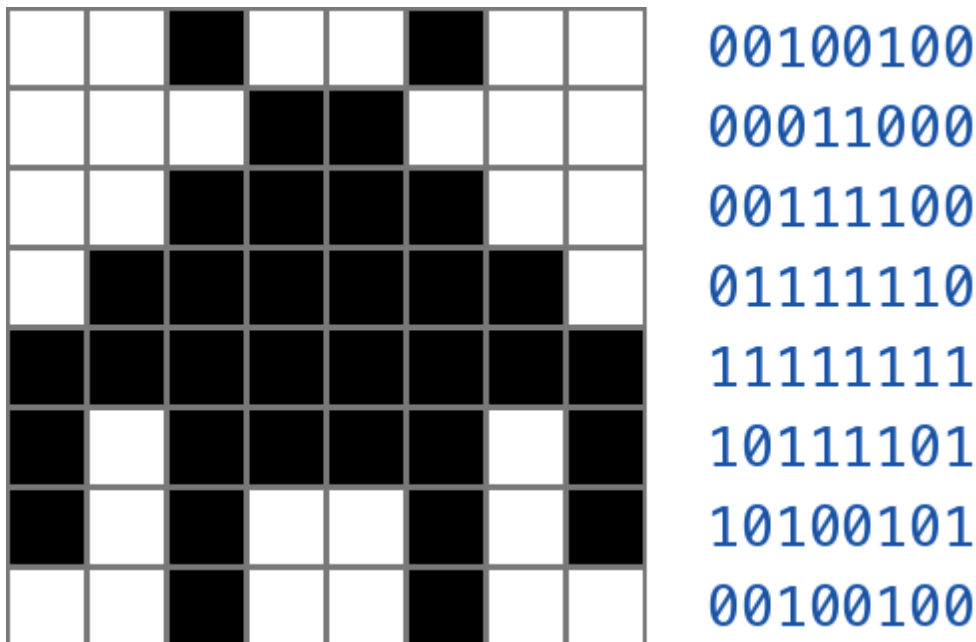
LO1 • A picture is a grid of pixels, and each pixel is a number

OCR 1.2.4 — representing images · 1-bit colour depth

Below is a picture 8 pixels wide and 8 pixels tall. Two things turn it into binary, and both of them are always given to you.

The index, which says what each pattern means. Here 0 is white and 1 is black.

The order, which starts at the **top left** and runs **left to right, top to bottom**, the way you read a page.



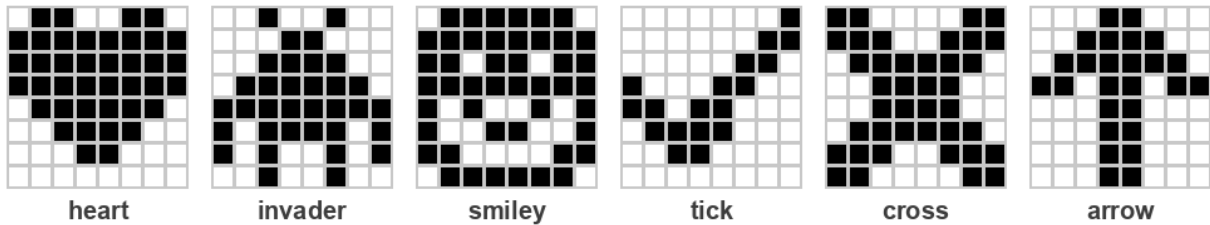
Two colours need only one bit per pixel, so this is a **colour depth of 1**. Every row is 8 bits, and there are 8 rows.

$8 \times 8 = 64$ pixels, 1 bit each, so 64 bits, which is $64 \div 8 = 8$ bytes.

LO2 • The same idea, six more times

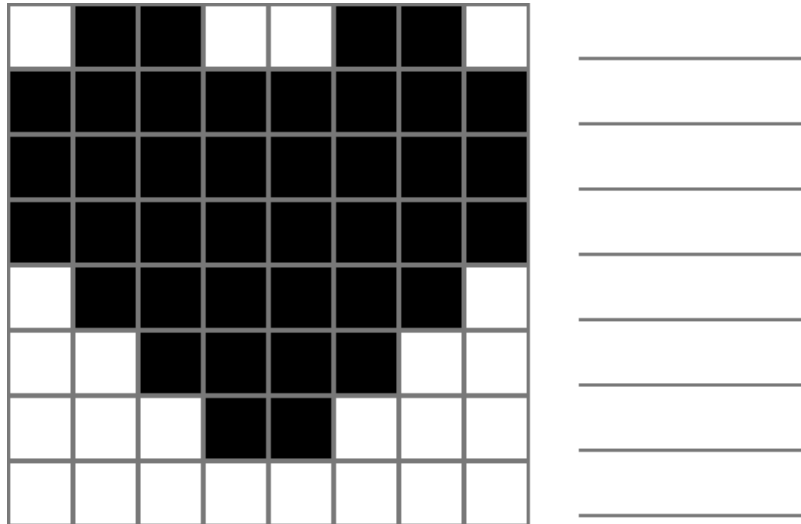
OCR 1.2.4 — encoding and decoding a bitmap

Every one of these is 8 by 8, at a colour depth of 1.



- (a) Write the binary for each row of this picture. The index is the same as before, 0 white and 1 black.

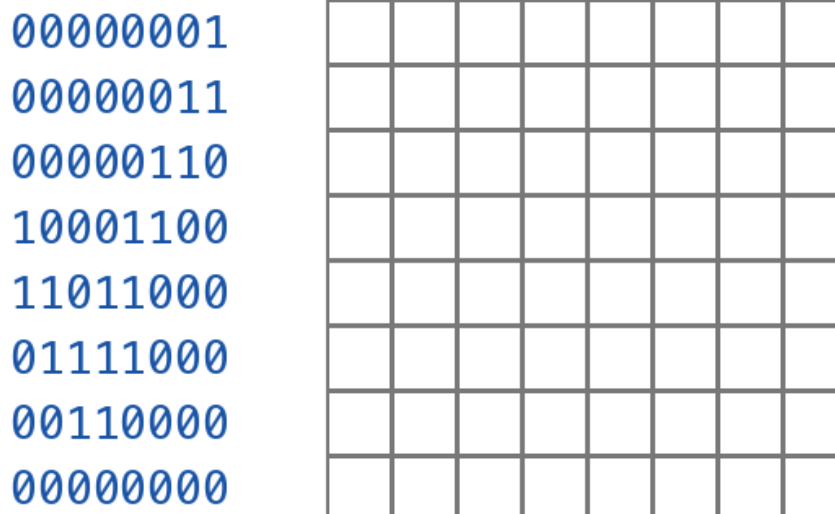
[4]



01100110 / 11111111 / 11111111 / 11111111 / 01111110 / 00111100 / 00011000 / 00000000

- (b) Now the other way round. Shade the grid to read this picture back out.

[4]



A tick.

Check: the shaded squares make a picture you recognise.

- (c) Both pictures above are 8 pixels by 8 pixels at a colour depth of 1. State how many bytes each one needs.

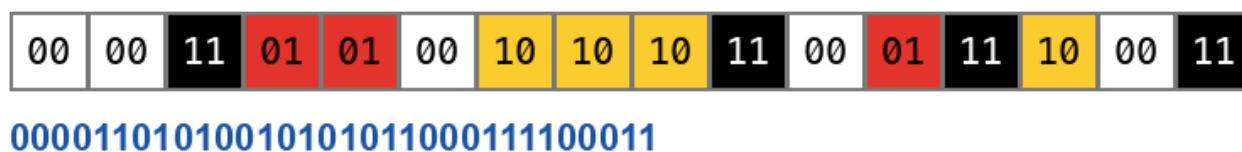
[2]

$8 \times 8 \times 1 = 64$ **bits, and** $64 \div 8 = 8$ **bytes.**

LO3 • More than two colours, so the index does more work

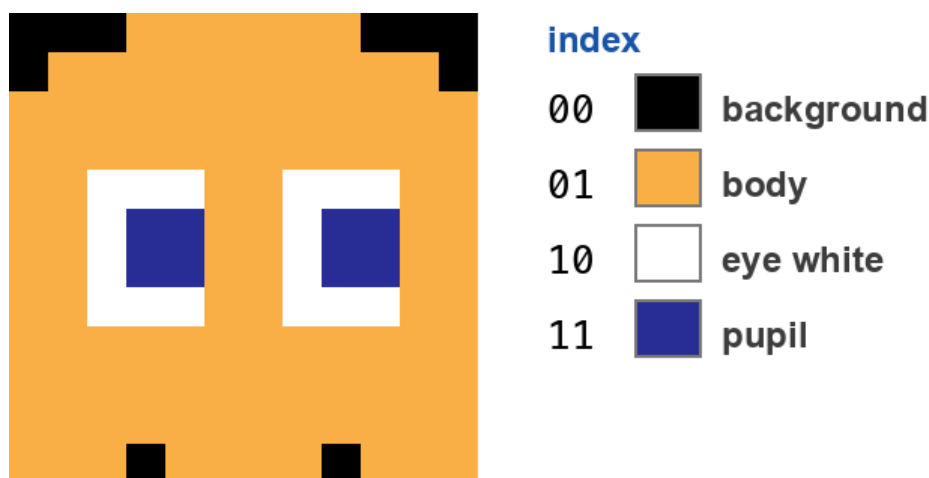
With four colours, one bit per pixel is no longer enough. Four colours need **2 bits** each, because $2^2 = 4$, and the **index** says which pattern means which colour. Here it is 00 white, 01 red, 10 yellow, 11 black.

Writing each pixel's code onto it turns the picture straight into the bits.

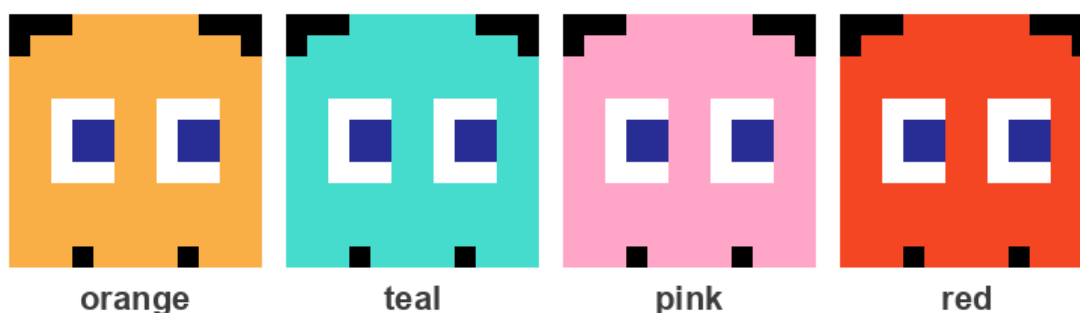


Sixteen pixels, 2 bits each, so 32 bits, which is $32 \div 8 = 4$ bytes.

The same thing works on a whole picture rather than one row. This ghost uses **exactly four colours** — background, body, eye white and pupil — so it too is a **colour depth of 2**.



Here is the part worth remembering. All four of these ghosts are stored with **exactly the same bits**. Only the index changed.



So the bits on their own do not tell you what the picture looks like. That is why an exam question always prints the index next to the data, and why reading it off is the first thing to do.

- (d) The ghost is 12 pixels wide and 12 pixels tall, at a colour depth of 2. Calculate its file size in bytes.

[3]

$12 \times 12 = 144$ **pixels**. $144 \times 2 = 288$ **bits**. $288 \div 8 = 36$ **bytes**.

- (e) A designer wants to add a fifth colour to the ghost. State the colour depth this now needs, and give the number of colours that depth makes available.

[2]

3 bits, which makes $2^3 = 8$ colours available. (Five colours do not fit in 2 bits, and depth goes up in whole bits, so the next step up is 3.)

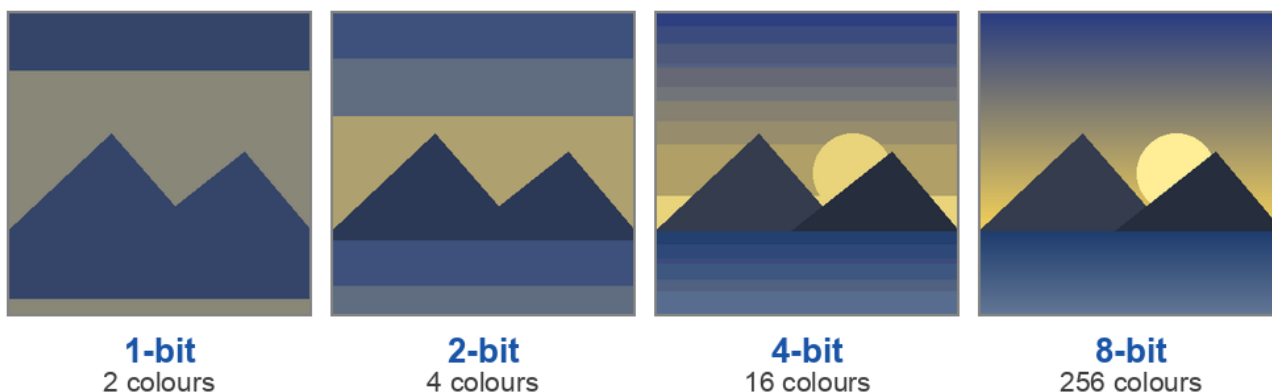
LO4 • More bits buy more colours

OCR 1.2.4 — colour depth and its effect on quality

Colour depth is the number of bits stored for one pixel. Those bits work like the place-value columns from Booklet 1, so n bits give 2^n colours. It is the same doubling table you already have.

Colour depth	1 bit	2 bits	3 bits	4 bits	8 bits
Colours available	$2^1 = 2$	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$	$2^8 = 256$

Here is one picture stored four times, at four different colour depths.



And here is the formula, worth writing down in exactly this form because it is short enough to remember under pressure.

$$\text{Size (B)} = \text{W} \times \text{H} \times \text{D} \div 8$$

- (f) Look along the four pictures above. Describe what more colour depth gives you, and say what it costs.

[3]

More colour depth means more colours available, so shades can be blended and fine detail and shadow shown, and the picture looks closer to the real thing. It costs storage: every extra bit is another bit stored for every single pixel, so the file gets bigger.

- (g) The 1-bit picture lost its sun completely, and the flat-coloured ghost on the previous page looks perfect at only 2 bits. Explain why the two pictures cope so differently with a low colour depth.

[3]

The ghost is made of a few flat areas of solid colour, and four colours is already all it contains, so nothing is lost. The sunset is made of shades blending smoothly into one another, and at a low colour depth many different shades have to be stored as the same colour, so the blends collapse into bands and small features disappear. The colour depth a picture needs depends on how many colours the picture actually contains.

- (h) An image is 200 pixels wide, 150 pixels tall, and has a colour depth of 8 bits. Calculate its file size in bytes.

[3]

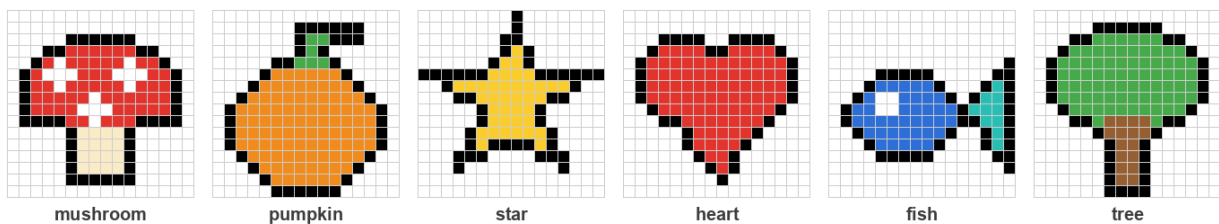
$200 \times 150 = 30,000$ **pixels**. $30,000 \times 8 = 240,000$ **bits**. $240,000 \div 8 = 30,000$ **bytes**. (One mark for the pixel count, one for the bit total, one for the conversion to bytes.)

L05 • Your turn

Piskel · piskelapp.com · free, in the browser, no account needed

Open piskelapp.com and choose **Create Sprite**. Set the canvas to **32 × 32**. (If it already opens at 32 by 32, you are ready to draw.)

Pick one subject and build it twice, once at each colour budget below. Keep the same subject both times, because the comparison is between the two budgets rather than between two different pictures. The third budget is shown to you at the end.



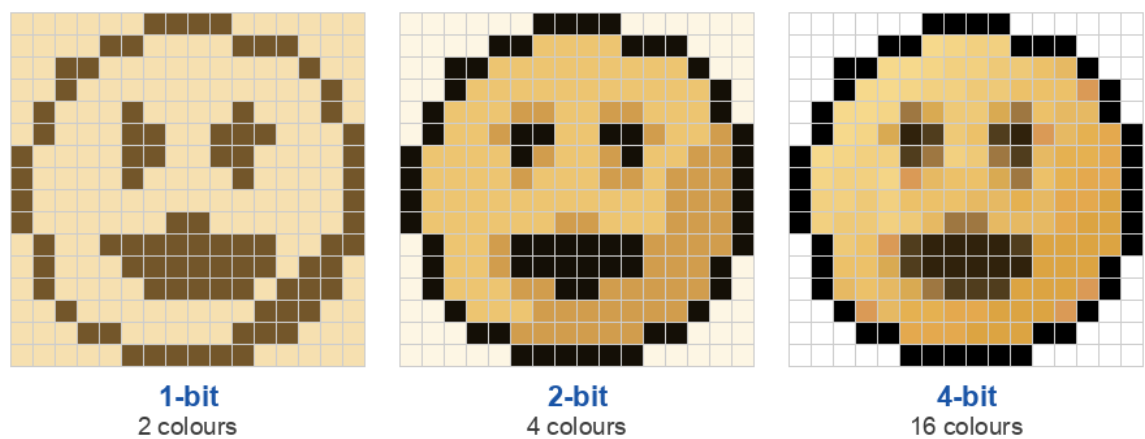
- (i) **Build 1 — 1-bit.** Use **exactly 2 colours**.

Check: count the swatches you have used. Two means a colour depth of 1.

- (j) **Build 2 — 2-bit.** Build it again with **exactly 4 colours**, then write your index here, the way the ghost's index was written.

Code	00	01	10	11
Colour				

(k) **Build 3 is done for you.** Sixteen colours takes a long while to choose and longer still to fill in, so here is the same picture already built at all three budgets.



Look along the three. Describe one thing the 16-colour version shows that the 4-colour version cannot.

[2]

Any one of: the face is shaded from light at the top left to darker at the bottom right, so it looks like a ball rather than a flat disc; the outline is a separate black from the shading, so the edge stays crisp; there are several yellows rather than one. Accept anything naming detail that needs more than four colours to store.

(l) Your two builds and the example above are all 32 by 32, so each one holds $32 \times 32 = 1024$ pixels. Complete the table.

[3]

Picture	Colour depth	Total bits	Size in bytes
Build 1	1 bit	1024	128
Build 2	2 bits	2048	256
The example	4 bits	4096	512

(m) Look down the final column. Describe what happens to the file size each time one bit is added to the colour depth.

[2]

It doubles. Each extra bit doubles the colours available and also doubles the bits stored for every pixel, so 128 bytes becomes 256, and 256 becomes 512.